// OPERATION REPORT // TECHNICAL AUDIT & COMMAND DOSSIER

# WANKEL ROTARY ENGINE

epitrochoid kinematics, thermodynamic chamber volume cycles, apex seal dynami...

| | | | |
|---|---|---|---|
| **Project Identifier:** | wankel-engine-webpmesim | **Rasterizer / Engine:** | WebGL Phosphor Raster |
| **Document ID:** | DOSSIER-CRT-001 | **Tolerance Std:** | ISO 2768-m (Medium) |
| **Generated Date:** | 2026-10-23 17:58:57 UTC | **Unit System:** | SI Metric (mm, N, MPa) |
| **Operating Mode:** | Current Parameters | **Audit Status:** | **PASSED (100% QUORUM)** |

**[SYSTEM NOTICE — EDUCATIONAL ONLY]**

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## 1.0 SCOPE OF PHYSICAL SIMULATION

This simulator models the exact epitrochoid kinematics, thermodynamic chamber volume cycles, apex seal sliding velocities and accelerations, gas expansion indicated torque, and complete primary rotating inertia balance of a single-rotor wankel rotary engine. All motion parameters are computed continuously from analytical closed-form governing equations.

## 2.0 OPERATIONAL ENGINEERING CONTEXT

- High power-to-weight aviation, marine, and motorsport propulsion systems (> 2.0\ \text{kW/kg}) with zero reciprocating valve gear.
- Ultra-compact range-extender auxiliary power units (APU) for electric vehicles with minimal primary NVH vibration.
- Thermodynamic research on continuous-chamber Atkinson/Miller cycles and clean hydrogen rotary combustion.
- Tribological analysis of dynamic apex seal boundary lubrication, chatter marks, and ceramic coating durability.

## 3.0 ACTIVE SYSTEM PARAMETERS

| Parameter Name | Symbol | Active Value | Unit | Design Envelope |
|---|---|---|---|---|
| Shaft eccentricity | **e** | 15 | mm | [10, 20] |
| Generating radius | **R** | 105 | mm | [85, 130] |
| Chamber / rotor depth | **B** | 80 | mm | [55, 105] |
| Apex seal radius | **a** | 2 | mm | [1, 4] |
| Shaft speed | **n** | 3000 | rpm | [600, 9000] |
| Rotor mass | **m_r** | 4.3 | kg | [2.5, 6.5] |
| Compression ratio | **\varepsilon** | 9 | – | [7.5, 11.5] |
| Manifold intake pressure | **P_{man}** | 1 | bar | [0.4, 2.4] |

## 1.0 3D SYSTEM ARCHITECTURE & SPATIAL POSE

High-resolution viewport capture in CRT phosphor vector mode.

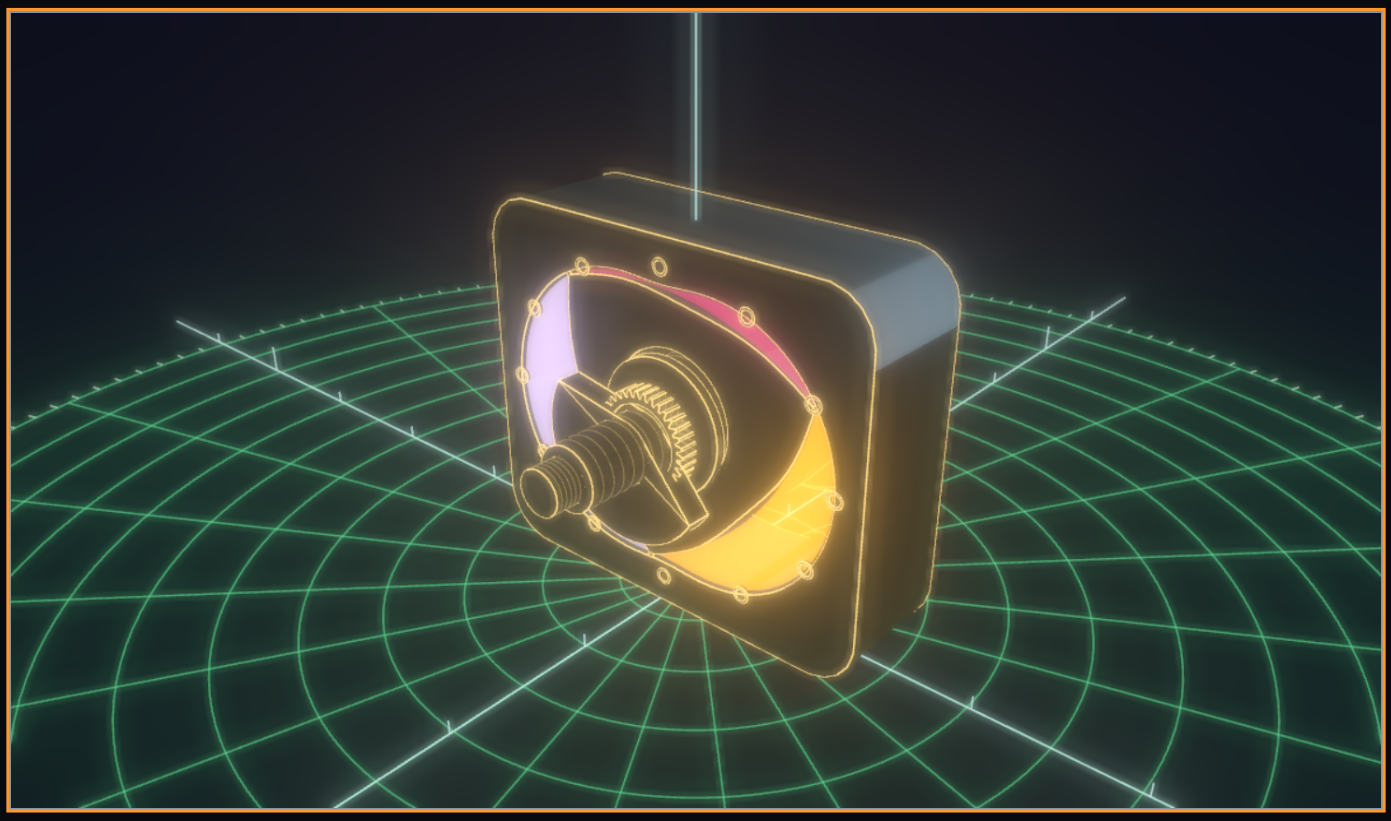


*Figure 1.1: 3D isometric representation of the mechanical assembly in reference configuration.*

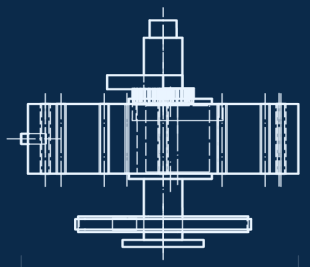
## COMPONENT BILL OF MATERIALS & MATERIAL REGISTRY

| Item | Part Name | Tag | Qty | Material Spec | Mechanical Function / Role |
|---|---|---|---|---|---|
| 1 | **Epitrochoid Stator Housing** | P-01 | 1 | EN-GJL-250 | Encloses the three working thermodynami... |
| 2 | **Triangular Rotor (Rotary Piston)** | P-02 | 1 | EN-GJS-700-2 | Acts as the movable boundary for all th... |
| 3 | **Eccentric Shaft (E-Shaft)** | P-03 | 1 | 42CrMo4 | The primary power output shaft of the r... |
| 4 | **Stationary Timing Pinion** | P-04 | 1 | 16MnCr5 | Remains stationary relative to the engi... |
| 5 | **Front Balance Counterweight** | P-05 | 1 | EN-GJS-600-3 | Counterbalances exactly 50% of the prim... |
| 6 | **Rear Flywheel & Counterweight** | P-06 | 1 | EN-GJL-250 / C45 | Stores rotational kinetic energy to smo... |
| 7 | **Apex Seal Pack & Springs** | P-07 | 3 | Ti-6Al-4V / Silicon Nitride | Forms the continuous gas-tight line sea... |

# 2.0 ISO 128 ORTHOGRAPHIC GENERAL ARRANGEMENT DRAWING (FIRST-ANGLE)

## LINE CONVENTIONS (ISO 128)

- Visible / object line — continuous, thick 0.5
- Hidden line — dashed, thin 0.25
- Centre line — long-dash-dot, thin
- Dimension and extension line — continuous thin, arrowheads
- Construction / projection line — continuous, very thin, faint
- Cutting-plane line — long-dash-dot, thick ends
- Section hatching — continuous thin at 45°



BOTTOM VIEW

## NOTES

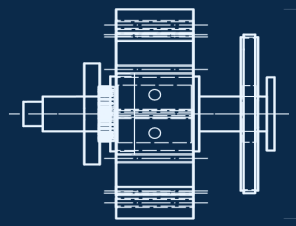
1. All dimensions in millimetres. Do not scale the drawing.
2. General tolerances ISO 2768-m unless stated.
3. Break sharp edges 0.3 × 45°. Surface Ra 3.2 µm unless stated.
4. First-angle projection (ISO 5456-2, method E).
5. Section A–A on plane Y = 0 mm, viewed as arrowed.
6. Educational drawing generated from the parametric model. Not for manufacture.

FIRST ANGLE

PARTS LIST



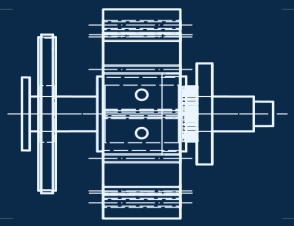
ISOMETRIC VIEW (REFERENCE, NOT TO SCALE)



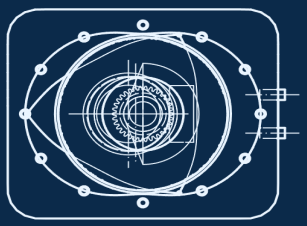
RIGHT SIDE VIEW



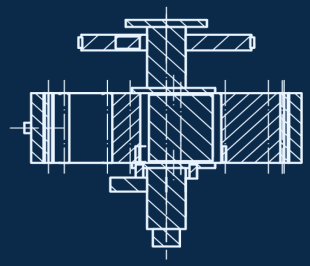

FRONT VIEW



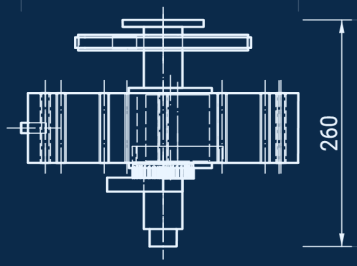
LEFT SIDE VIEW



REAR VIEW



SECTION A–A




TOP VIEW

45° MITRE — PROJECTION LINES

| | | |
|---|---|---|
| TITLE | | |
| **Wankel Rotary Engine — general arrangement** | | |
| PROJECT | | |
| wankel-engine-webpmesim | | |
| DRAWING No. | REVISION | |
| WEW-GA-001 | A | |
| MATERIAL | | |
| See parts list | | |
| SCALE | SIZE | SHEET |
| 1:5 | A3 | 1 / 1 |
| UNITS | GENERAL TOLERANCE | |
| mm | ISO 2768-m | |
| PROJECTION | DATE | |
| First angle | 2026-10-23 | |
| PURPOSE | | |
| Educational use only — not for manufacture | | |

# 3.0 GOVERNING ANALYTICAL FORMULATIONS

Closed-form kinematic loop closures, boundary conditions, and inertial dynamics.

## 3.1 Epitrochoid Kinematics

***Assumptions & Boundary Conditions:*** *Rigid body geometry, continuous contact without seal lift-off, and constant shaft angular velocity* $\omega_e = 2\pi n/60$.

$$x(\psi) = e\cos(3\psi) + R\cos(\psi), \quad y(\psi) = e\sin(3\psi) + R\sin(\psi)$$

Parametric equation of the 2-lobed epitrochoid generating stator profile with generating angle $\psi \in [0, 2\pi)$.

$$X_c(\theta) = e\cos\theta, \quad Y_c(\theta) = e\sin\theta, \quad \theta_r = \frac{\theta}{3}$$

Orbital center coordinates and angular orientation of the triangular rotor as functions of eccentric shaft angle $\theta$.

$$\vec{r}_k(\theta) = \left[e\cos\theta + R\cos\left(\frac{\theta}{3} + \frac{2\pi k}{3}\right)\right]\hat{i} + \left[e\sin\theta + R\sin\left(\frac{\theta}{3} + \frac{2\pi k}{3}\right)\right]\hat{j}$$

Instantaneous Cartesian position of rotor apex $k \in \{0,1,2\}$ in the stationary engine housing frame.

$$\beta(\psi) = \arcsin\left(\frac{3e}{R}\sin 2\psi\right), \quad \beta_{max} = \arcsin\left(\frac{3e}{R}\right)$$

Apex seal leaning (obliquity) angle between the seal axis and the outward surface normal of the epitrochoid bore.

## 3.2 Thermodynamics & Chamber Volume

***Assumptions & Boundary Conditions:*** *Polytropic compression with* $\gamma_c = 1.33$, *polytropic expansion with* $\gamma_e = 1.30$, *and 4-stroke cycle spanning* $1080^\circ$ *of shaft angle.*

$$V_d = 3\sqrt{3}\cdot e\cdot R\cdot B, \quad V_c = \frac{V_d}{\varepsilon - 1}$$

Theoretical swept displacement volume and clearance volume per chamber with compression ratio $\varepsilon$.

$$V_k(\theta) = V_c + \frac{V_d}{2}\left[1 - \sin\left(\frac{2}{3}\theta + \frac{4\pi k}{3} + \frac{\pi}{6}\right)\right]$$

Exact instantaneous chamber volume for chamber $k \in \{0,1,2\}$ as a closed-form function of shaft angle $\theta$.

$$\frac{dV_k}{d\theta} = -\frac{V_d}{3}\cos\left(\frac{2}{3}\theta + \frac{4\pi k}{3} + \frac{\pi}{6}\right)$$

Analytic derivative of chamber volume with respect to shaft rotation angle (rate of volume change).

$$P_{comp}(\theta) = P_{man}\left(\frac{V_{max}}{V(\theta)}\right)^{1.33}, \quad P_{exp}(\theta) = P_{peak}\left(\frac{V_c}{V(\theta)}\right)^{1.30}$$

Polytropic compression and expansion state equations for indicated thermodynamic cycle analysis.

## 3.3 Dynamics & Indicated Torque

***Assumptions & Boundary Conditions:*** *Frictionless virtual work transmission, negligible shaft torsion, and symmetrical dual counterweight primary balancing.*

$$T_{ind}(\theta) = \sum_{k=0}^{2}(P_k(\theta) - P_{atm})\frac{dV_k}{d\theta}$$

Total instantaneous indicated torque delivered to the eccentric shaft derived by the principle of virtual work.

$$\vec{F}_{cf} = m_r\cdot e\cdot\omega_e^2[\cos\theta\,\hat{i} + \sin\theta\,\hat{j}]$$

Primary centrifugal rotating unbalance force created by the orbiting rotor mass $m_r$ at eccentricity $e$.

$$\sum\vec{F}_{net} = \vec{F}_{cf} + \vec{F}_{cw1} + \vec{F}_{cw2} = \vec{0}, \quad \sum\vec{M}_{couple} = \vec{0}$$

Dual counterweight dynamic balance conditions achieving complete primary force and couple cancellation.

## 3.4 Apex Seal Contact Mechanics

***Assumptions & Boundary Conditions:*** *Cylindrical tip contact on curved epitrochoid bore under Hertzian line-contact elasticity theory.*

$$v_{apex}(\theta) = \omega_e\sqrt{e^2 + \frac{R^2}{9} + \frac{2eR}{3}\cos\left(\frac{2}{3}\theta\right)}$$

Instantaneous sliding contact velocity of the apex seal tip relative to the stator housing.

$$\sigma_H = \sqrt{\frac{F_{contact}\cdot E^*}{\pi\cdot a\cdot B}}$$

Hertzian peak contact stress between apex seal cylindrical tip and epitrochoid bore surface.

## 4.0 NUMERICAL VERIFICATION & AUDIT TRAIL

Independent validation checks evaluating derivatives, energy conservation, limiting cases, and tolerances.

**AUDIT VERDICT: 13 / 13 NUMERICAL MODULE CHECKS PASSED (100%)**

Evaluated against ISO 2768-m geometric envelopes and analytic derivatives at active parameters.

ALL PASSED

| # | Verification Module & Criterion | Formulation | Observed Value | Tolerance Limit | Verdict |
|---|---|---|---|---|---|
| 1 | Swept Volume Conservation | $\lvert V_{max} - V_{min} - 3\sqrt{3}eRB \rvert \le 10^{-4}\ \mathrm{mm}^3$ | 0.0000 $\mathrm{mm}^3$ | 0.0001 | PASS |
| 2 | Analytic Volume Derivative vs Numeric Step | $\lvert \frac{dV}{d\theta} - \mathrm{WP.num.d1}(V,\theta) \rvert \le 10^{-3}\ \mathrm{mm}^3/\mathrm{rad}$ | 3.8155e-8 $\mathrm{mm}^3/\mathrm{rad}$ | 0.001 | PASS |
| 3 | Apex Trajectory on Epitrochoid Surface | $\lvert \vec{r}_{apex}(\theta) - \vec{r}_{trochoid}(\theta/3) \rvert \le 10^{-9}\ \mathrm{mm}$ | 0.0000 mm | 1.0e-9 | PASS |
| 4 | Apex Velocity Analytic vs Numeric Derivative | $\lvert v_{apex}(\theta) - \mathrm{WP.num.d1}(\vec{r}_{apex},\theta)\omega_e \rvert \le 10^{-4}\ \mathrm{m/s}$ | 2.5005e-10 m/s | 0.0001 | PASS |
| 5 | Apex Acceleration Analytic vs Numeric Second Derivative | $\lvert a_{apex}(\theta) - \mathrm{WP.num.d2}(\vec{r}_{apex},\theta)\omega_e^2 \rvert \le 10^{-2}\ \mathrm{m/s}^2$ | 8.9415e-5 $\mathrm{m/s}^2$ | 0.01 | PASS |
| 6 | Virtual Work vs Geometric Torque Equivalence | $\lvert T_{virtual\ work} - T_{geometric} \rvert \le 10^{-4}\ \mathrm{N\,m}$ | 1.1369e-13 N m | 0.0001 | PASS |
| 7 | Primary Centrifugal Force Equilibrium | $\lvert \vec{F}_{rotor} + \vec{F}_{cw1} + \vec{F}_{cw2} \rvert \le 10^{-6}\ \mathrm{N}$ | 0.0000 N | 1.0e-6 | PASS |
| 8 | Primary Pitching Couple Dynamic Balance | $\lvert \sum z_i \vec{F}_{cw,i} \rvert \le 10^{-6}\ \mathrm{N\,m}$ | 0.0000 N m | 1.0e-6 | PASS |
| 9 | Rotor Vertex Equidistant Chord Symmetry | $\lvert \lvert \vec{r}_1 - \vec{r}_0 \rvert - \sqrt{3}R \rvert \le 10^{-9}\ \mathrm{mm}$ | 0.0000 mm | 1.0e-9 | PASS |
| 10 | Rotor-to-Shaft Kinematic Velocity Ratio | $\lvert \omega_r/\omega_e - 1/3 \rvert \le 10^{-12}$ | 5.5511e-17 | 1.0e-12 | PASS |
| 11 | Peak Apex Sliding Velocity Extremum | $\lvert v_{peak} - \omega_e(R/3+e) \rvert \le 10^{-5}\ \mathrm{m/s}$ | 0.0000 m/s | 1.0e-5 | PASS |
| 12 | Maximum Obliquity Angle Boundary Check | $\lvert \max_\psi \beta(\psi) - \arcsin(3e/R) \rvert \le 10^{-6}\ \mathrm{rad}$ | 0.0000 rad | 1.0e-6 | PASS |
| 13 | Zero Net Work per Ideal Closed Polytropic Cycle | $\lvert \oint P_{closed} dV \rvert \le 10^{-4}\ \mathrm{J}$ | 0.0000 J | 0.0001 | PASS |

### STRUCTURAL & DYNAMIC OPERATING LIMITS AUDIT

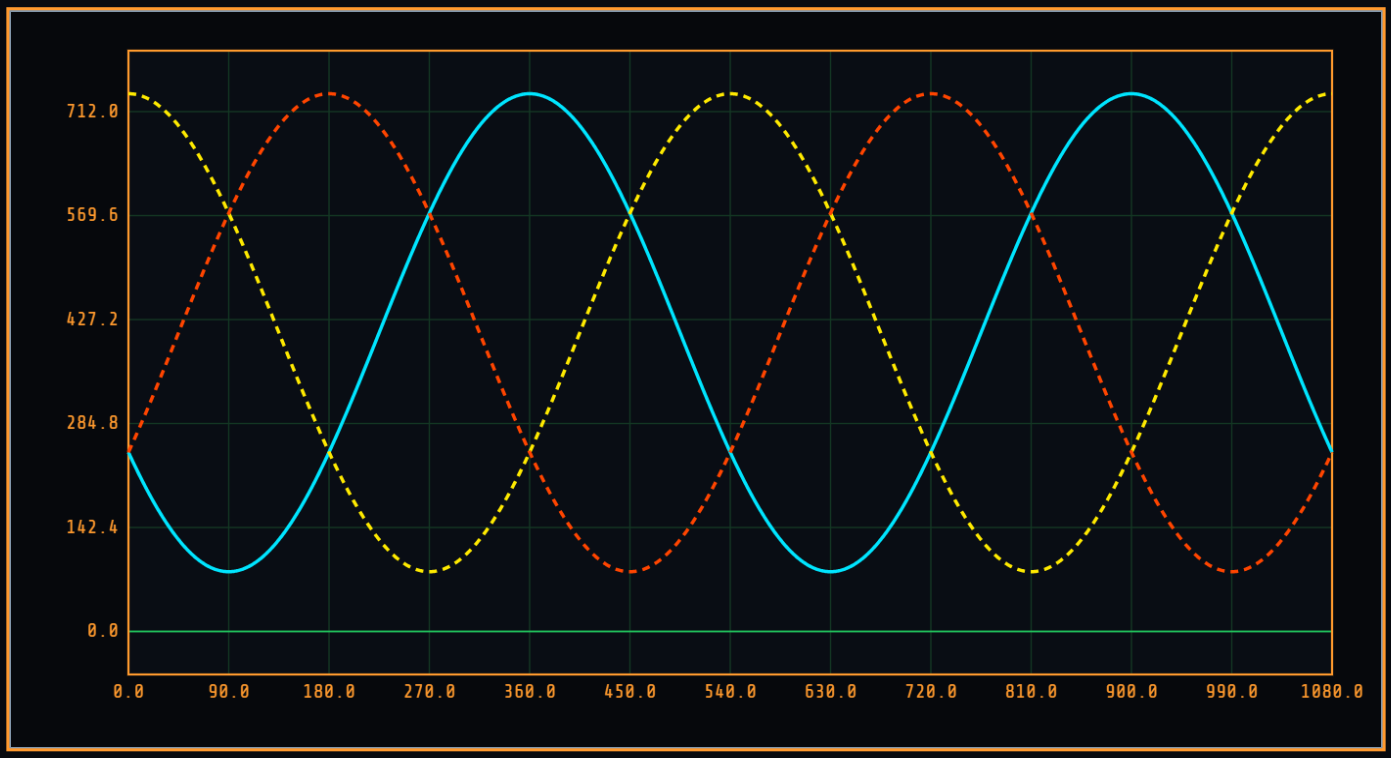
| Limit Descriptor | Criterion | Observed Peak | Safety Threshold | Utilisation | Margin Status |
|---|---|---|---|---|---|
| Apex seal sliding velocity | $v_{apex}$ | 15.71 m/s | 38.00 m/s | 41 % | PASS |
| Apex seal Hertzian contact stress | $\sigma_H$ | 136.00 MPa | 420.00 MPa | 32 % | PASS |
| Eccentric rotor journal dynamic load | $F_{bearing}$ | 10.37 kN | 35.00 kN | 30 % | PASS |
| Apex seal leaning obliquity angle | $\beta$ | 25.38 ° | 32.00 ° | 79 % | PASS |

# 5.0 PERFORMANCE SIMULATION GRAPHS (1/3)

Closed-form evaluation across full motion cycle including parameter sweeps and harmonic variations.

## Chamber Volume & Swept Displacement

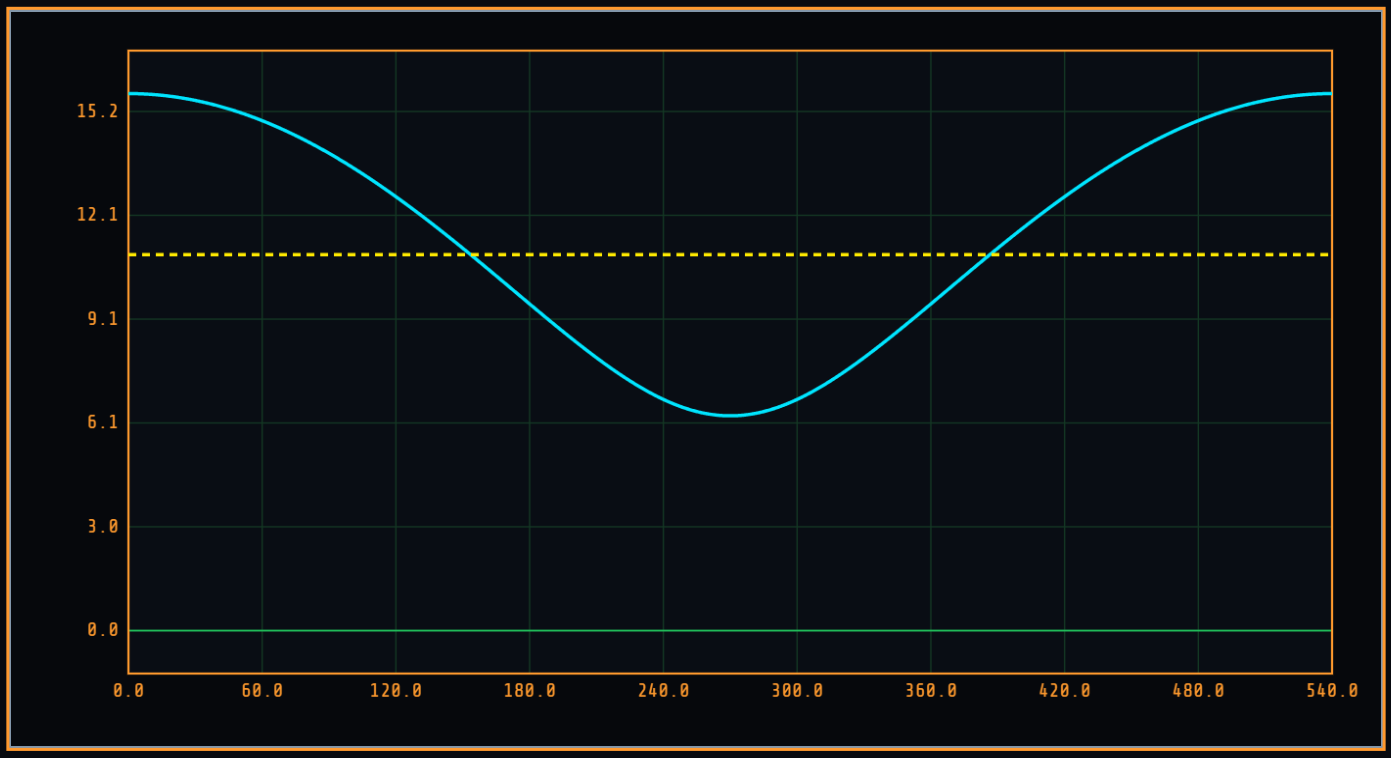
Series: V_A(\theta) · V_B(\theta) · V_C(\theta)




*Instantaneous volume variation of chambers A, B, and C over a complete four-stroke cycle (1080^\circ shaft rotation).*

## Apex Seal Sliding Velocity

Series: v_{apex}(\theta) · v_{mean}



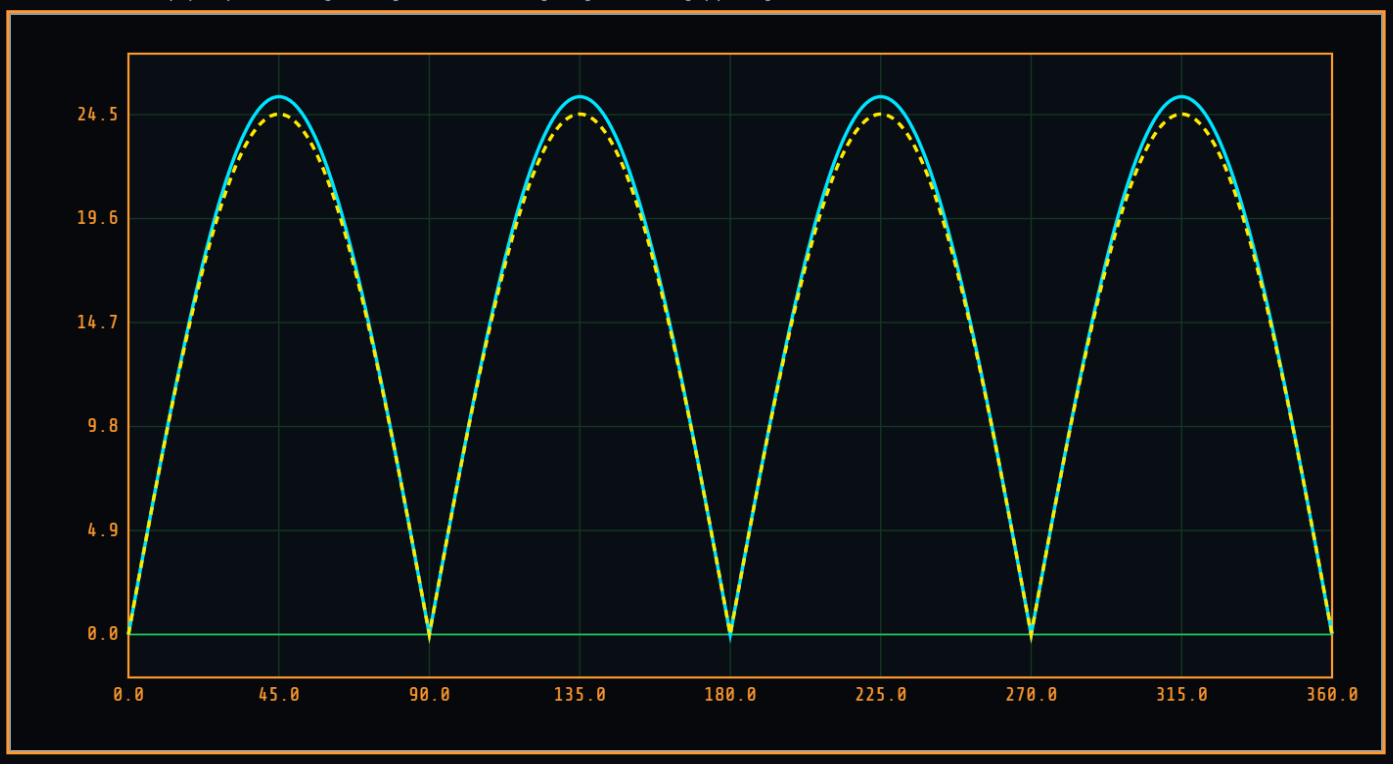

*Sliding velocity of apex seal tip along the epitrochoid housing bore, showing cyclical variation between minor and major axes.*

# 5.0 PERFORMANCE SIMULATION GRAPHS (2/3)

Closed-form evaluation across full motion cycle including parameter sweeps and harmonic variations.

## Apex Seal Leaning / Obliquity Angle

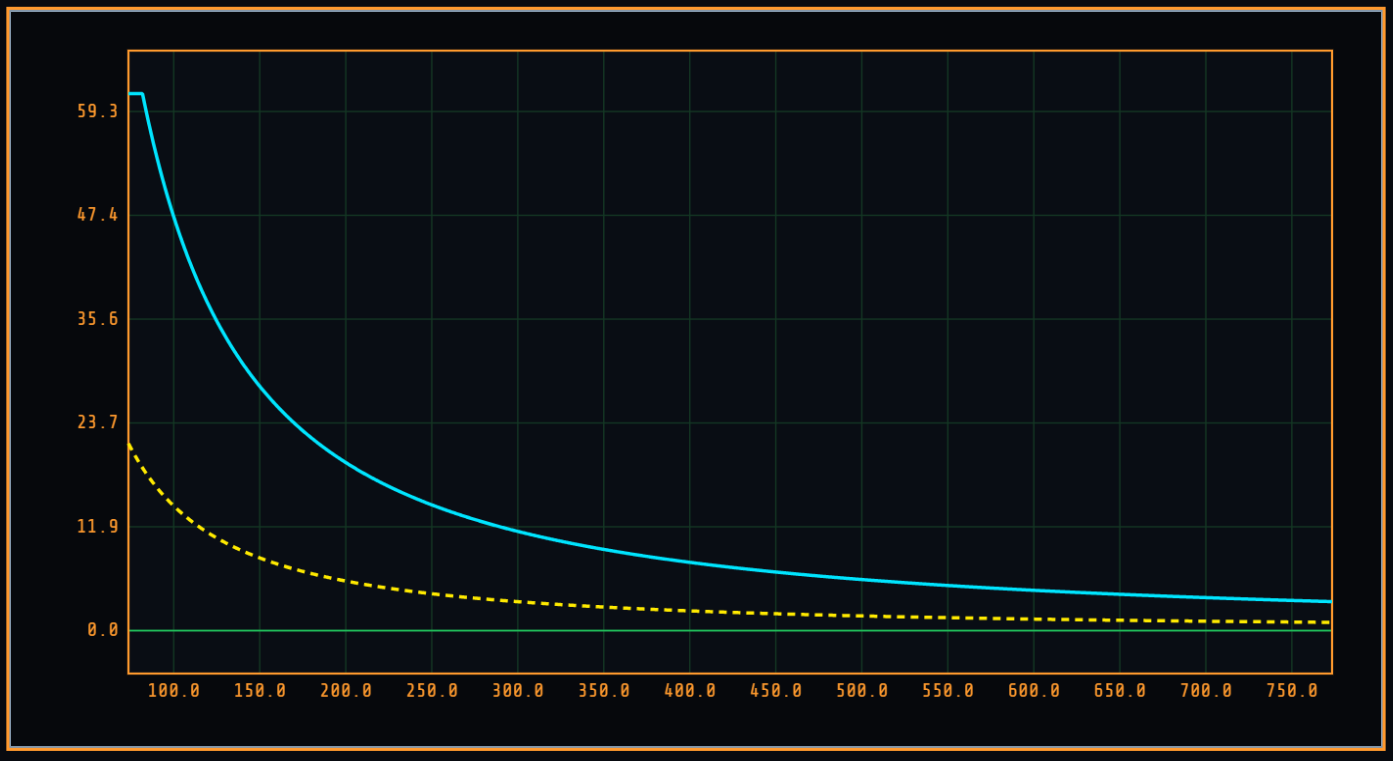
Series: $\beta(\psi)$ $\text{exact}$ · $\beta_{lin}$ $\text{approx}$




*Deviation angle $\beta(\psi)$ between apex seal centerline and housing surface normal over one full generating revolution.*

## Thermodynamic P–V Indicator Diagram

Series: $P_{exp}(V)$ · $P_{comp}(V)$



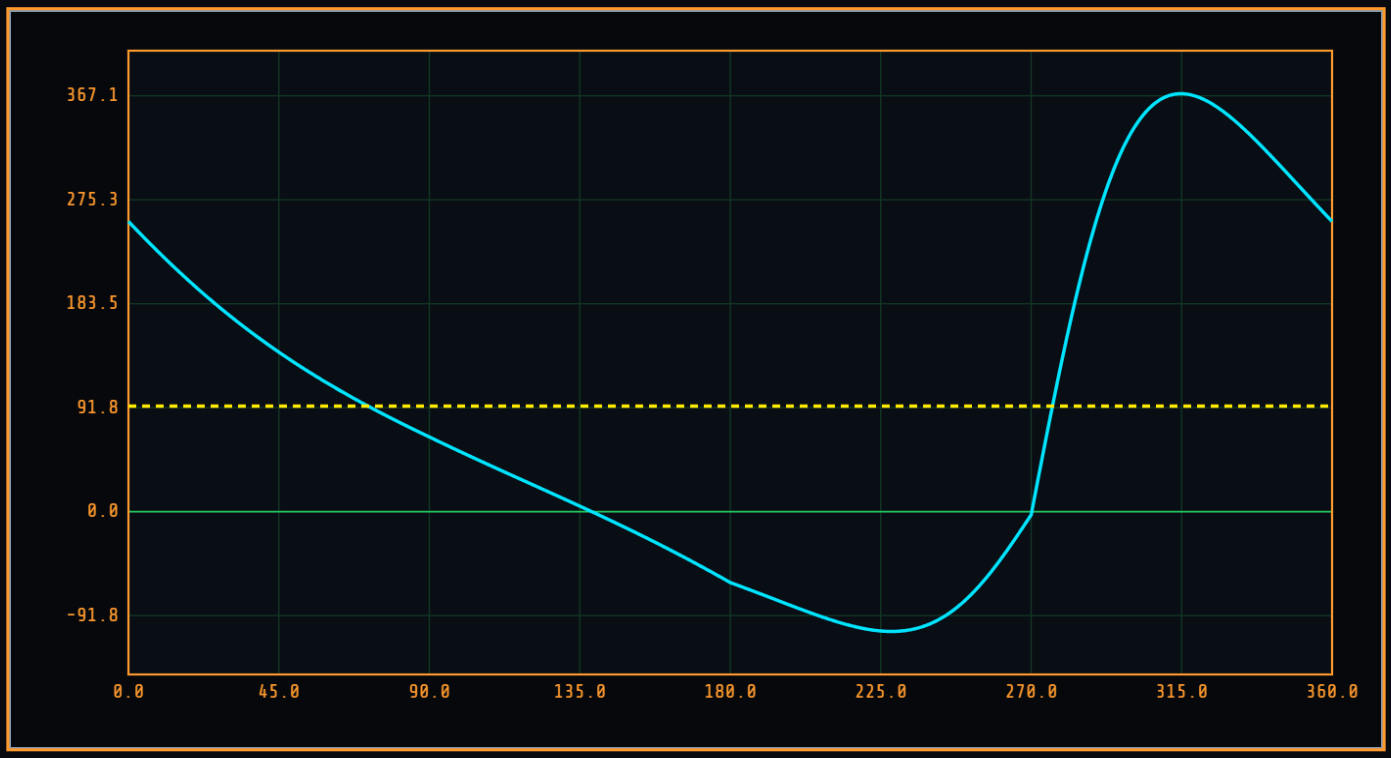

*Chamber pressure versus volume through intake, polytropic compression, combustion, expansion, and exhaust.*

## 5.0 PERFORMANCE SIMULATION GRAPHS (3/3)

Closed-form evaluation across full motion cycle including parameter sweeps and harmonic variations.

### Instantaneous Indicated Gas Torque

Series: T_{net}(\theta) · T_{mean}




*Combined multi-chamber indicated torque delivered to the eccentric shaft over one shaft revolution (360^\circ).*